

数理统计-知识总结（仅数一三）

1. 抽样分布：
$$\left\{ \begin{array}{l} \chi^2 \text{分布:} \begin{cases} \text{条件: } X_1, X_2, \dots, X_n, \text{ 相互独立, 且同服从 } N(0,1) \\ \text{结论: } \chi^2 = X_1^2 + X_2^2 + \dots + X_n^2 \sim \chi^2(n) \end{cases} \\ t \text{分布:} \begin{cases} \text{条件: } X \sim N(0,1), Y \sim \chi^2(n), \text{ 且 } X, Y \text{ 相互独立} \\ \text{结论: } T = \frac{X}{\sqrt{Y/n}} \sim t(n) \end{cases} \\ F \text{分布:} \begin{cases} \text{条件: } X \sim \chi^2(n_1), Y \sim \chi^2(n_2), \text{ 且 } X, Y \text{ 相互独立} \\ \text{结论: } F = \frac{X/n_1}{Y/n_2} \sim F(n_1, n_2) \end{cases} \end{array} \right.$$

2. 统计量：
$$\left\{ \begin{array}{l} \text{设 } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \text{ 则} \\ \textcircled{1} \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \text{ 或 } \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \\ \textcircled{2} \frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi^2(n-1) \\ \textcircled{3} \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \sim \chi^2(n) \\ \textcircled{4} \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1) \end{array} \right.$$

3. 点估计：
$$\left\{ \begin{array}{l} \text{矩估计: } E(X^n) = A_n = \frac{1}{n} \sum_{i=1}^n X_i^n \Rightarrow \theta = \bar{\theta}(X_1, X_2, \dots, X_n) \\ \text{极大似然估计:} \begin{cases} \text{似然函数: 连续型: } L(\theta) = \prod_{i=1}^n f(x_i; \theta); \text{ 离散型: } L(\theta) = \prod_{i=1}^n P(x_i; \theta) \\ \text{求最值:} \begin{cases} \text{有驻点: } \ln L(\theta) = \sum_{i=1}^n P(x_i; \theta), \frac{d \ln L(\theta)}{d \theta} = 0 \Rightarrow \hat{\theta} = \hat{\theta}(x_1, x_2, \dots, x_n) \\ \text{无驻点: 根据实际意义求出最大值点} \end{cases} \end{cases} \\ \text{最大似然估计的不变性: } \hat{\theta} \text{ 是 } \theta \text{ 的最大似然估计, 函数 } \varphi(\theta) \text{ 具有单值反函数} \\ \text{则 } \varphi(\hat{\theta}) \text{ 是 } \varphi(\theta) \text{ 的最大似然估计, 即: } \hat{\varphi}(\theta) = \varphi(\hat{\theta}). \end{array} \right.$$

4. 无偏性/有效性/一致性 (仅数一):
$$\left\{ \begin{array}{l} \text{无偏性: } E(\hat{\theta}) = \theta \\ \text{有效性: } \hat{\theta}_1 \text{ 与 } \hat{\theta}_2 \text{ 都是 } \theta \text{ 的无偏估计量,} \\ \quad \text{若 } D(\hat{\theta}_1) < D(\hat{\theta}_2), \text{ 则称 } \hat{\theta}_1 \text{ 较 } \hat{\theta}_2 \text{ 有效} \\ \text{一致性: 当 } n \rightarrow \infty \text{ 时, } \hat{\theta} \text{ 依概率收敛于 } \theta \\ \quad \text{则称 } \hat{\theta} \text{ 为 } \theta \text{ 的一致估计量} \end{array} \right.$$

5. 区间估计 (仅数一):

待估参数	分布	置信区间
$\mu(\sigma^2 \text{ 已知})$	$U = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$(\bar{X} - \mu_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + \mu_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}})$ $P\{ U \geq \mu_{\frac{\alpha}{2}}\} = \alpha$
$\mu(\sigma^2 \text{ 未知})$	$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$	$(\bar{X} - t_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}})$ $P\{ T \geq t_{\frac{\alpha}{2}}\} = \alpha$

6. 假设检验 (仅数一):

常用结论:

原假设 H_0	统计量及分布	H_0 的拒绝域
$\mu = \mu_0$ (σ^2 已知)	$U = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$ u = \left \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right \geq u_{\frac{\alpha}{2}}$
$\mu = \mu_0$ (σ^2 未知)	$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$	$ t = \left \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \right \geq t_{\frac{\alpha}{2}}(n-1)$

两类错误:
$$\left\{ \begin{array}{l} \text{弃真: 假设本来成立, 但被错误地否认了} \\ \text{存伪: 把本来不成立的假设错误地接受了} \end{array} \right.$$